```
                                                                  ___  ____  ____  ____  ____(R)
                                                                 /__    /   ____/   /   ____/
                                                                ___/   /   /___/   /   /___/
                                                                  Statistics/Data analysis

                                                                  User: Stata Results 10.11.2025

  ___  ____  ____  ____  ____ ®
 /__    /   ____/   /   ____/      StataNow 19.5
___/   /   /___/   /   /___/       SE—Standard Edition

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Notes:
      1. Unicode is supported; see help unicode advice.
      2. Maximum number of variables is set to 5,000 but can be increased;
          see help set maxvar.

 1 . import excel "C:\Users\DELL\OneDrive\FMT EXCEL (Source File) (final).xlsx", sheet("SUMMARY") firstro
   > w
   (6 vars, 310 obs)

 2 . do "C:\Users\DELL\AppData\Local\Temp\STD48c_000000.tmp"

 3 .
 4 .
 5 . gen daily_date = date(DATE, "MDY")   // Convert Excel text data into Stata daily date format
   (1 missing value generated)

 6 . format daily_date %td // Display the date in daily format (dd/mm/yyyy)

 7 . gen mdate = mofd(daily_date)
   (1 missing value generated)

 8 . format mdate %tm

 9 . tsset mdate // Declare the data as monthly time series

   Time variable: mdate, 2000m1 to 2025m9
           Delta: 1 month

10 .
11 . rename INFLATION inf // Rename var Inflation as inf

12 . rename BOEBASERATE intrate // Rename var BOE BASE RATE  as intrate

13 . rename UNEMPLOYMENTRATE Unemp // Rename var Unemployment rate as Unemp

14 . rename BRENTCRUDEOIL Cop // Rename var Brent Crude Oil Price as Cop
```

```
15 .
16 . gen l_Cop = ln(Cop) // oil price (logged)
   (1 missing value generated)

17 .
18 . tsline inf intrate Unemp l_Cop // Plot time series graphs for all variables

19 .
20 . hist inf // Show distribution of Inflation
   (bin=17, start=-.1, width=.65882353)

21 . hist intrate // Show distribution of BOE BASE RATE
   (bin=17, start=.1, width=.34705882)

22 . hist Unemp // Show distribution of Exchange rate
   (bin=17, start=3.6, width=.28823529)

23 . hist l_Cop // Show distribution of Brent Crude Oil Price
   (bin=17, start=2.9517803, width=.11697923)

24 .
25 . scatter intrate Unemp l_Cop inf // Scatter plot of variables to see relationships

26 .
27 . dfuller inf // ADF Test for Inflation

   Dickey-Fuller test for unit root          Number of obs  = 308
   Variable: inf                             Number of lags =   0

   H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                     Test      ──────── critical value ─────────
                statistic           1%           5%          10%
   ──────────────────────────────────────────────────────────────
    Z(t)            -1.673       -3.455       -2.878       -2.570
   ──────────────────────────────────────────────────────────────
   MacKinnon approximate p-value for Z(t) = 0.4449.

28 . dfuller intrate // ADF Test for BOE BASE RATE

   Dickey-Fuller test for unit root          Number of obs  = 308
   Variable: intrate                         Number of lags =   0

   H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                     Test      ──────── critical value ─────────
                statistic           1%           5%          10%
   ──────────────────────────────────────────────────────────────
    Z(t)            -1.341       -3.455       -2.878       -2.570
   ──────────────────────────────────────────────────────────────
   MacKinnon approximate p-value for Z(t) = 0.6104.
```

```
29 . dfuller Unemp // ADF Test for Unemployment rate

   Dickey-Fuller test for unit root          Number of obs  = 308
   Variable: Unemp                           Number of lags =   0

   H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                     Test      ──────── critical value ─────────
                statistic          1%           5%          10%
   ──────────────────────────────────────────────────────────────
    Z(t)            -0.704      -3.455       -2.878       -2.570
   ──────────────────────────────────────────────────────────────
   MacKinnon approximate p-value for Z(t) = 0.8456.

30 . dfuller l_Cop // ADF Test for Crude Oil Price

   Dickey-Fuller test for unit root          Number of obs  = 308
   Variable: l_Cop                           Number of lags =   0

   H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                     Test      ──────── critical value ─────────
                statistic          1%           5%          10%
   ──────────────────────────────────────────────────────────────
    Z(t)            -2.296      -3.455       -2.878       -2.570
   ──────────────────────────────────────────────────────────────
   MacKinnon approximate p-value for Z(t) = 0.1733.

31 .
32 . dfuller D.inf // ADF test on Inflation

   Dickey-Fuller test for unit root          Number of obs  = 307
   Variable: D.inf                           Number of lags =   0

   H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                     Test      ──────── critical value ─────────
                statistic          1%           5%          10%
   ──────────────────────────────────────────────────────────────
    Z(t)           -14.397      -3.455       -2.878       -2.570
   ──────────────────────────────────────────────────────────────
   MacKinnon approximate p-value for Z(t) = 0.0000.

33 . dfuller D.intrate // ADF test on differenced BOE BASE RATE

   Dickey-Fuller test for unit root          Number of obs  = 307
   Variable: D.intrate                       Number of lags =   0

   H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                     Test      ──────── critical value ─────────
                statistic          1%           5%          10%
   ──────────────────────────────────────────────────────────────
    Z(t)           -10.055      -3.455       -2.878       -2.570
   ──────────────────────────────────────────────────────────────
   MacKinnon approximate p-value for Z(t) = 0.0000.
```

```
34 . dfuller D.Unemp // ADF test on differenced Unemployment rate

   Dickey-Fuller test for unit root          Number of obs   = 307
   Variable: D.Unemp                         Number of lags =   0

   H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                     Test        ---------- critical value ----------
                statistic            1%           5%          10%
   ---------------------------------------------------------------
    Z(t)           -11.883       -3.455       -2.878       -2.570
   ---------------------------------------------------------------
   MacKinnon approximate p-value for Z(t) = 0.0000.

35 . dfuller D.l_Cop // ADF test on differenced Crude Oil Price

   Dickey-Fuller test for unit root          Number of obs   = 307
   Variable: D.l_Cop                         Number of lags =   0

   H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                     Test        ---------- critical value ----------
                statistic            1%           5%          10%
   ---------------------------------------------------------------
    Z(t)           -14.978       -3.455       -2.878       -2.570
   ---------------------------------------------------------------
   MacKinnon approximate p-value for Z(t) = 0.0000.

36 .
37 . varsoc inf intrate Unemp l_Cop, maxlag(12) // Determine optimal VAR lag length for Variables

   Lag-order selection criteria

      Sample: 2001m1 thru 2025m9                              Number of obs = 297
```

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-1880.39				3.81147	12.6895	12.7094	12.7393
1	519.743	4800.3	16	0.000	4.1e-07	-3.36528	-3.2657	-3.11654
2	588.718	137.95	16	0.000	2.8e-07	-3.722	-3.54276	-3.27428*
3	616.964	56.492	16	0.000	2.6e-07	-3.80447	-3.54557	-3.15776
4	651.83	69.733	16	0.000	2.3e-07	-3.93152	-3.59295*	-3.08582
5	676.218	48.775	16	0.000	2.2e-07	-3.988	-3.56977	-2.94331
6	687.757	23.079	16	0.112	2.2e-07	-3.95796	-3.46007	-2.71428
7	712.204	48.894	16	0.000	2.1e-07*	-4.01484*	-3.43729	-2.57217
8	722.049	19.689	16	0.235	2.2e-07	-3.97339	-3.31618	-2.33173
9	735.655	27.212	16	0.039	2.3e-07	-3.95727	-3.22039	-2.11662
10	744.82	18.331	16	0.305	2.4e-07	-3.91125	-3.09471	-1.87161
11	756.335	23.029	16	0.113	2.4e-07	-3.88104	-2.98484	-1.64242
12	770.627	28.584*	16	0.027	2.5e-07	-3.86954	-2.89368	-1.43192

```
   * optimal lag
   Endogenous: inf intrate Unemp l_Cop
    Exogenous: _cons
```

```
38 .
39 . vecrank inf intrate Unemp l_Cop, lags(7) trend(constant)  // Determine number of cointegration relat
   > ionships

   Johansen tests for cointegration
   Trend: Constant                          Number of obs  = 302
   Sample: 2000m8 thru 2025m9               Number of lags =   7
   ──────────────────────────────────────────────────────────────────────────────
                                                          Critical
   Maximum                                       Trace      value
      rank  Params           LL  Eigenvalue   statistic         5%
         0     100    691.9413           .     62.6376      47.21
         1     107   710.06811     0.11312     26.3839*     29.68
         2     112   717.81532     0.05001     10.8895      15.41
         3     115   721.74926     0.02572      3.0216       3.76
         4     116   723.26009     0.00996
   ──────────────────────────────────────────────────────────────────────────────
   * selected rank

40 .
41 . vec inf intrate Unemp l_Cop, lags(7) rank(1) trend(constant) // Estimate VEC

   Vector error-correction model

   Sample: 2000m8 thru 2025m9                      Number of obs     =        302
                                                   AIC               =  -3.993829
   Log likelihood =  710.0681                      HQIC              =  -3.467832
   Det(Sigma_ml)  =  1.07e-07                      SBIC              =  -2.679207

   Equation           Parms      RMSE     R-sq      chi2     P>chi2
   ────────────────────────────────────────────────────────────────
   D_inf                 26     .329629   0.3296   135.6684   0.0000
   D_intrate             26     .134506   0.4495   225.3277   0.0000
   D_Unemp               26     .092835   0.2900   112.7079   0.0000
   D_l_Cop               26     .100752   0.1018    31.2652   0.2185
   ────────────────────────────────────────────────────────────────

   ──────────────────────────────────────────────────────────────────────────────
                  | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
   ───────────────┼──────────────────────────────────────────────────────────────
   D_inf          |
           _ce1   |
            L1.   |  -.0487869   .0146909    -3.32   0.001    -.0775806   -.0199932
                  |
            inf   |
            LD.   |   .0290919   .0579901     0.50   0.616    -.0845666    .1427504
           L2D.   |   .0399323   .0571247     0.70   0.485      -.07203    .1518945
           L3D.   |    .184024   .0562053     3.27   0.001     .0738637    .2941844
           L4D.   |   .1670916   .0568911     2.94   0.003     .0555871    .2785961
           L5D.   |    .087077    .055734     1.56   0.118    -.0221597    .1963136
           L6D.   |    .206953   .0564539     3.67   0.000     .0963054    .3176005
                  |
        intrate   |
            LD.   |     .42712   .1495968     2.86   0.004     .1339157    .7203243
           L2D.   |  -.3128314   .1583205    -1.98   0.048    -.6231339    -.002529
           L3D.   |  -.0256193    .157276    -0.16   0.871    -.3338745    .2826359
           L4D.   |   -.412037   .1556202    -2.65   0.008     -.717047    -.107027
           L5D.   |    -.03829   .1601888    -0.24   0.811    -.3522543    .2756743
           L6D.   |   .2910205    .151847     1.92   0.055     -.006594    .5886351
                  |
          Unemp   |
            LD.   |   .1534436   .2165074     0.71   0.478     -.270903    .5777903
           L2D.   |  -.4299039    .218291    -1.97   0.049    -.8577464   -.0020614
           L3D.   |  -.0501737    .219845    -0.23   0.819     -.481062    .3807146
           L4D.   |  -.4488533   .2184594    -2.05   0.040    -.8770258   -.0206808
```

L5D.	-.1363229	.2147016	-0.63	0.525	-.5571304	.2844845
L6D.	.5041147	.2130205	2.37	0.018	.0866022	.9216271
l_Cop						
LD.	.6411303	.2050077	3.13	0.002	.2393225	1.042938
L2D.	.3068351	.2098793	1.46	0.144	-.1045208	.718191
L3D.	.0290881	.2089057	0.14	0.889	-.3803596	.4385357
L4D.	.1392411	.206065	0.68	0.499	-.2646389	.5431211
L5D.	.4121835	.2051179	2.01	0.044	.0101598	.8142072
L6D.	.0251106	.2056172	0.12	0.903	-.3778917	.4281129
_cons	-.0017909	.0192119	-0.09	0.926	-.0394455	.0358637
D_intrate						
_ce1						
L1.	.0194296	.0059947	3.24	0.001	.0076803	.031179
inf						
LD.	.0231991	.0236631	0.98	0.327	-.0231796	.0695778
L2D.	-.0116869	.0233099	-0.50	0.616	-.0573734	.0339997
L3D.	-.035656	.0229348	-1.55	0.120	-.0806073	.0092953
L4D.	.0090893	.0232146	0.39	0.695	-.0364105	.0545891
L5D.	-.0079336	.0227425	-0.35	0.727	-.052508	.0366408
L6D.	-.0235763	.0230362	-1.02	0.306	-.0687264	.0215738
intrate						
LD.	.3323688	.0610435	5.44	0.000	.2127258	.4520118
L2D.	.0642581	.0646032	0.99	0.320	-.0623619	.1908781
L3D.	.2691756	.064177	4.19	0.000	.1433909	.3949602
L4D.	-.1266563	.0635014	-1.99	0.046	-.2511167	-.0021959
L5D.	-.1374969	.0653656	-2.10	0.035	-.2656112	-.0093827
L6D.	.1495439	.0619617	2.41	0.016	.0281012	.2709866
Unemp						
LD.	-.0143973	.0883466	-0.16	0.871	-.1875535	.1587588
L2D.	-.1365492	.0890744	-1.53	0.125	-.3111318	.0380335
L3D.	.0341806	.0897085	0.38	0.703	-.1416449	.2100061
L4D.	-.1145469	.0891431	-1.28	0.199	-.2892642	.0601704
L5D.	-.0305329	.0876098	-0.35	0.727	-.2022448	.1411791
L6D.	-.0125119	.0869238	-0.14	0.886	-.1828793	.1578555
l_Cop						
LD.	.2508538	.0836541	3.00	0.003	.0868947	.4148128
L2D.	.0969744	.085642	1.13	0.257	-.0708808	.2648296
L3D.	-.0367277	.0852447	-0.43	0.667	-.2038042	.1303489
L4D.	.183131	.0840856	2.18	0.029	.0183263	.3479357
L5D.	.0000672	.0836991	0.00	0.999	-.1639799	.1641144
L6D.	-.1306042	.0839028	-1.56	0.120	-.2950507	.0338423
_cons	-.0049993	.0078395	-0.64	0.524	-.0203643	.0103658
D_Unemp						
_ce1						
L1.	.0110262	.0041375	2.66	0.008	.002917	.0191355
inf						
LD.	-.0074384	.016332	-0.46	0.649	-.0394485	.0245717
L2D.	.0253068	.0160882	1.57	0.116	-.0062256	.0568391
L3D.	-.0137727	.0158293	-0.87	0.384	-.0447975	.0172522
L4D.	.0139342	.0160225	0.87	0.384	-.0174692	.0453377
L5D.	-.0139036	.0156966	-0.89	0.376	-.0446683	.0168612
L6D.	-.0197243	.0158993	-1.24	0.215	-.0508865	.0114378
intrate						
LD.	.0126827	.0421315	0.30	0.763	-.0698936	.095259

L2D.	-.0420332	.0445884	-0.94	0.346	-.1294249	.0453585
L3D.	-.0506657	.0442943	-1.14	0.253	-.1374808	.0361495
L4D.	-.1118311	.0438279	-2.55	0.011	-.1977322	-.0259299
L5D.	.0044921	.0451146	0.10	0.921	-.0839309	.0929151
L6D.	.0291167	.0427653	0.68	0.496	-.0547017	.112935
Unemp						
LD.	.194283	.0609758	3.19	0.001	.0747725	.3137934
L2D.	.172107	.0614782	2.80	0.005	.051612	.292602
L3D.	-.0393974	.0619158	-0.64	0.525	-.1607502	.0819554
L4D.	.0232961	.0615256	0.38	0.705	-.0972918	.143884
L5D.	.0917073	.0604673	1.52	0.129	-.0268064	.2102209
L6D.	.0323848	.0599938	0.54	0.589	-.0852009	.1499705
l_Cop						
LD.	-.0314893	.0577371	-0.55	0.585	-.144652	.0816734
L2D.	.0095861	.0591092	0.16	0.871	-.1062657	.125438
L3D.	.0276801	.0588349	0.47	0.638	-.0876342	.1429945
L4D.	.0150915	.0580349	0.26	0.795	-.0986548	.1288379
L5D.	-.0258137	.0577682	-0.45	0.655	-.1390373	.0874098
L6D.	-.1121835	.0579088	-1.94	0.053	-.2256826	.0013157
_cons	-.0002212	.0054107	-0.04	0.967	-.010826	.0103836
D_l_Cop						
_ce1						
L1.	.0034841	.0044903	0.78	0.438	-.0053168	.012285
inf						
LD.	.0043943	.0177248	0.25	0.804	-.0303458	.0391344
L2D.	.0040947	.0174603	0.23	0.815	-.0301269	.0383163
L3D.	.0001022	.0171793	0.01	0.995	-.0335686	.0337731
L4D.	-.0028172	.0173889	-0.16	0.871	-.0368989	.0312645
L5D.	-.0230993	.0170353	-1.36	0.175	-.0564878	.0102892
L6D.	-.0119973	.0172553	-0.70	0.487	-.0458171	.0218224
intrate						
LD.	.1228724	.0457247	2.69	0.007	.0332537	.2124912
L2D.	-.0388315	.0483911	-0.80	0.422	-.1336764	.0560134
L3D.	-.0143583	.0480719	-0.30	0.765	-.1085774	.0798609
L4D.	-.0941063	.0475658	-1.98	0.048	-.1873335	-.0008791
L5D.	.0289795	.0489622	0.59	0.554	-.0669846	.1249437
L6D.	-.0371513	.0464125	-0.80	0.423	-.1281181	.0538155
Unemp						
LD.	-.0353381	.0661761	-0.53	0.593	-.165041	.0943647
L2D.	-.0927463	.0667213	-1.39	0.165	-.2235177	.038025
L3D.	.0119921	.0671963	0.18	0.858	-.1197102	.1436944
L4D.	.0502817	.0667728	0.75	0.451	-.0805905	.1811539
L5D.	.1133114	.0656242	1.73	0.084	-.0153096	.2419325
L6D.	-.0174377	.0651103	-0.27	0.789	-.1450516	.1101763
l_Cop						
LD.	.1046605	.0626612	1.67	0.095	-.0181532	.2274743
L2D.	-.0839918	.0641502	-1.31	0.190	-.209724	.0417404
L3D.	-.0712882	.0638527	-1.12	0.264	-.1964371	.0538607
L4D.	-.0426013	.0629844	-0.68	0.499	-.1660485	.0808458
L5D.	-.007722	.0626949	-0.12	0.902	-.1306017	.1151578
L6D.	.073101	.0628475	1.16	0.245	-.0500779	.1962799
_cons	.0035016	.0058722	0.60	0.551	-.0080076	.0150109

Cointegrating equations

Equation	Parms	chi2	P>chi2
_ce1	3	16.69765	0.0008

Identification: beta is exactly identified

Johansen normalization restriction imposed

beta	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
_ce1						
inf	1	.	.	.	.	.
intrate	-.1161359	.1187219	-0.98	0.328	-.3488265	.1165547
Unemp	.0097891	.1960865	0.05	0.960	-.3745333	.3941115
l_Cop	-2.139254	.5660445	-3.78	0.000	-3.248681	-1.029827
_cons	6.469164	.	.	.	.	.

```
42 .
43 . veclmar // Test VECM residuals for autocorrelation using LM test
```

Lagrange-multiplier test

lag	chi2	df	Prob > chi2
1	17.5629	16	0.35011
2	15.9530	16	0.45624

H0: no autocorrelation at lag order

```
44 .
45 . vecstable, graph // Check VECM stability using eigenvalue stability condition
```

Eigenvalue stability condition

Eigenvalue	Modulus
1	1
1	1
1	1
.9342603 + .1062877i	.940287
.9342603 - .1062877i	.940287
-.335846 + .7725581i	.8424
-.335846 - .7725581i	.8424
.8179245	.817925
-.4371696 + .6609112i	.792415
-.4371696 - .6609112i	.792415
.187471 + .7505508i	.77361
.187471 - .7505508i	.77361
.3972794 + .6553906i	.766399
.3972794 - .6553906i	.766399
.5671984 + .5082485i	.761597
.5671984 - .5082485i	.761597
.7442083	.744208
-.7413066 + .01927172i	.741557
-.7413066 - .01927172i	.741557
.6787754	.678775
-.5205993 + .4311143i	.675931
-.5205993 - .4311143i	.675931
-.6395554	.639555
.3516877 + .4973962i	.609169
.3516877 - .4973962i	.609169
-.2678351 + .5438873i	.606258
-.2678351 - .5438873i	.606258

```
│   -.269618                │   .269618   │
└──────────────────────────┴─────────────┘
   The VECM specification imposes 3 unit moduli.

46 .
   end of do-file

47 . save "C:\Users\DELL\OneDrive\FMT STATA (output) (final).dta"
   file C:\Users\DELL\OneDrive\FMT STATA (output) (final).dta saved

48 .
```